ISLAMIC UNIVERSITY OF TECHNOLOGY

DHAKA, BANGLADESH

ORGANIZATION OF ISLAMIC COOPERATION

DATABASE MANAGEMENT SYSTEMS LAB

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Report preparing for Lab Task

Here is the problem statement -

Here, the underlined attributes denote the primary keys and the arcs denote the foreign key rela-

tionships.

1. Create a view named Advisor_Selection that shows the ID, name and department name of instructors.

```
CREATE OR REPLACE VIEW Advisor_Selection AS
SELECT ID , name , dept_name
FROM instructor ;
```

```
SQL> connect lab06/cse
Connected.
SQL> CREATE OR REPLACE VIEW Advisor_Selection AS
  2  SELECT ID , name , dept_name
  3  FROM instructor ;

View created.
```

2. Create another view named Student_Count using Advisor_Selection and advisor to show the name of the instructor and the number students assigned under them.

```
CREATE OR REPLACE VIEW Student_Count AS
SELECT a.name ,COUNT(d.s_id)as st_count
FROM Advisor_Selection a
Join advisor d ON a.id=d.i_id
group by a.name;
GROUP BY a.name;
```

```
SQL> CREATE OR REPLACE VIEW Student_Count AS
  2  SELECT a.name ,COUNT(d.s_id)as st_count
  3  FROM Advisor_Selection a
  4  Join advisor d ON a.id=d.i_id
  5  group by a.name;

View created.
```

3. Four categories of users have been identified in the database:

(a) Students should be able to view information regarding advisors and courses.

```
CREATE ROLE std_role;
GRANT SELECT ON advisor TO std_role;
GRANT SELECT ON course TO std_role;
```

```
SQL> CREATE ROLE std_role;

Role created.

SQL> GRANT SELECT ON advisor TO std_role;

Grant succeeded.

SQL> GRANT SELECT ON course TO std_role;

Grant succeeded.
```

(b) Course teachers should be able to view information about the students and courses.

```
CREATE ROLE t_role;
GRANT SELECT ON student TO t_role;
GRANT SELECT ON course TO t_role;
```

```
SQL> CREATE ROLE t_role;

Role created.

SQL> GRANT SELECT ON student TO t_role;

Grant succeeded.

SQL> GRANT SELECT ON course TO t_role;

Grant succeeded.
```

(c) Head of the Departments should have all the privileges that a course teacher has. Additionally, s/he should be able to add new instructors.

```
CREATE ROLE hod_role;
GRANT t_role TO hod_role;
GRANT SELECT,INSERT ON instructor TO hod_role;
```

```
SQL> CREATE ROLE hod_role;

Role created.

SQL> GRANT t_role TO hod_role;

Grant succeeded.

SQL> GRANT SELECT,INSERT ON instructor TO hod_role;

Grant succeeded.
```

(d) Administrator should be able to see information about department and instructors. They should also be able to update the department budget.

```
CREATE ROLE admin_role;
Grant select ON department TO admin_role;
Grant update(Budget) ON department TO admin_role;
Grant select ON instructor TO admin_role;
```

```
SQL> CREATE ROLE admin_role;

Role created.

SQL> Grant select ON department TO admin_role;

Grant succeeded.

SQL> Grant update(Budget) ON department TO admin_role;

Grant succeeded.

SQL> Grant select ON instructor TO admin_role;

Grant succeeded.
```

4. Create users under these roles and write relevant SQL queries to demonstrate that the imposed access control works.

```
drop user stud ;
drop user teach ;
drop user dept ;
drop user admin ;


create user stud identified by s;
create user teach identified by t ;
create user dept identified by d ;
create user admin identified by a ;
```

```
grant  create  session , resource , create tablespace
to stud;
grant  create  session , resource , create tablespace
to teach;
grant  create  session , resource , create tablespace
to dept;
grant  create  session , resource , create tablespace
to admin;


grant std_role to stud ;
grant t_role to teach ;
grant hod_role to dept ;
grant admin_role to admin ;

SELECT * FROM Advisor_Selection;

SELECT * FROM student;

SELECT * FROM course;


INSERT INTO instructor (ID, name, dept_name, salary)
VALUES ('12', 'Farhan', 'Comp. Sci.', 90000);
SELECT * FROM instructor;

UPDATE department SET budget = 130000 WHERE dept_name
= 'Comp. Sci.';

SELECT * FROM department;
SELECT * FROM instructor;
```

```
SQL> drop user stud ;

User dropped.

SQL> drop user teach ;

User dropped.

SQL> drop user dept ;

User dropped.

SQL> drop user admin ;

User dropped.

SQL>
SQL>
SQL> create user stud identified by s;

User created.

SQL> create user teach identified by t ;

User created.

SQL> create user dept identified by d ;

User created.

SQL> create user admin identified by a ;

User created.
```

```
SQL> grant  create  session , resource , create tablespace to stud;

Grant succeeded.

SQL> grant  create  session , resource , create tablespace to teach;

Grant succeeded.

SQL> grant  create  session , resource , create tablespace to dept;

Grant succeeded.

SQL> grant  create  session , resource , create tablespace to admin;

Grant succeeded.

SQL>
SQL>
SQL> grant std_role to stud ;

Grant succeeded.

SQL> grant t_role to teach ;

Grant succeeded.

SQL> grant hod_role to dept ;

Grant succeeded.

SQL> grant admin_role to admin ;

Grant succeeded.
```

```
SQL> SELECT * FROM Advisor_Selection;

ID    NAME                 DEPT_NAME
----- -------------------- --------------------
10101 Srinivasan           Comp. Sci.
12121 Wu                   Finance
15151 Mozart               Music
22222 Einstein             Physics
32343 El Said              History
33456 Gold                 Physics
45565 Katz                 Comp. Sci.
58583 Califieri            History
76543 Singh                Finance
76766 Crick                Biology
83821 Brandt               Comp. Sci.

ID    NAME                 DEPT_NAME
----- -------------------- --------------------
98345 Kim                  Elec. Eng.

12 rows selected.

SQL> SELECT * FROM course;

COURSE_I TITLE                                          DEPT_NAME
-------- ---------------------------------------------- ----------
----
   CREDITS
----------
BIO-101  Intro. to Biology                              Biology
         4

BIO-301  Genetics                                       Biology
         4

BIO-399  Computational Biology                          Biology
         3
```

```
COURSE_I TITLE                                              DEPT_NAME
-------- -------------------------------------------------- ----------
----
   CREDITS
----------
CS-101   Intro. to Computer Science                         Comp. Sci.
         4

CS-190   Game Design                                        Comp. Sci.
         4

CS-315   Robotics                                           Comp. Sci.
         3


COURSE_I TITLE                                              DEPT_NAME
-------- -------------------------------------------------- ----------
----
   CREDITS
----------
CS-319   Image Processing                                   Comp. Sci.
         3

CS-347   Database System Concepts                           Comp. Sci.
         3

EE-181   Intro. to Digital Systems                          Elec. Eng.
         3
```

```
COURSE_I TITLE                                              DEPT_NAME
-------- -------------------------------------------------- ----------------
----
   CREDITS
----------
FIN-201  Investment Banking                                 Finance
         3

HIS-351  World History                                      History
         3

MU-199   Music Video Production                             Music
         3


COURSE_I TITLE                                              DEPT_NAME
-------- -------------------------------------------------- ----------------
----
   CREDITS
----------
PHY-101  Physical Principles                                Physics
         4


13 rows selected.
```

```
SQL> SELECT * FROM student;

ID    NAME                 DEPT_NAME             TOT_CRED
----- -------------------- -------------------- ----------
00128 Zhang                Comp. Sci.                 102
12345 Shankar              Comp. Sci.                  32
19991 Brandt               History                     80
23121 Chavez               Finance                    110
44553 Peltier              Physics                     56
45678 Levy                 Physics                     46
54321 Williams             Comp. Sci.                  54
55739 Sanchez              Music                       38
70557 Snow                 Physics                      0
76543 Brown                Comp. Sci.                  58
76653 Aoi                  Elec. Eng.                  60

ID    NAME                 DEPT_NAME             TOT_CRED
----- -------------------- -------------------- ----------
98765 Bourikas             Elec. Eng.                  98
98988 Tanaka               Biology                    120

13 rows selected.
```

```
SQL> INSERT INTO instructor (ID, name, dept_name, salary) VALUES ('12', 'Far
han', 'Comp. Sci.', 90000);

1 row created.

SQL> SELECT * FROM instructor;

ID    NAME                 DEPT_NAME               SALARY
----- -------------------- -------------------- ----------
10101 Srinivasan           Comp. Sci.                65000
12121 Wu                   Finance                   90000
15151 Mozart               Music                     40000
22222 Einstein             Physics                   95000
32343 El Said              History                   60000
33456 Gold                 Physics                   87000
45565 Katz                 Comp. Sci.                75000
58583 Califieri            History                   62000
76543 Singh                Finance                   80000
76766 Crick                Biology                   72000
83821 Brandt               Comp. Sci.                92000

ID    NAME                 DEPT_NAME               SALARY
----- -------------------- -------------------- ----------
98345 Kim                  Elec. Eng.                80000
12    Farhan               Comp. Sci.                90000

13 rows selected.
```

```
SQL> UPDATE department SET budget = 130000 WHERE dept_name = 'Comp. Sci.';

1 row updated.

SQL>
SQL> SELECT * FROM department;

DEPT_NAME            BUILDING            BUDGET
-------------------- --------------- ----------
Biology              Watson               90000
Comp. Sci.           Taylor              130000
Elec. Eng.           Taylor               85000
Finance              Painter             120000
History              Painter              50000
Music                Packard              80000
Physics              Watson               70000

7 rows selected.
```

```
SQL> SELECT * FROM instructor;

ID    NAME                 DEPT_NAME                SALARY
----- -------------------- -------------------- ----------
10101 Srinivasan           Comp. Sci.                65000
12121 Wu                   Finance                   90000
15151 Mozart               Music                     40000
22222 Einstein             Physics                   95000
32343 El Said              History                   60000
33456 Gold                 Physics                   87000
45565 Katz                 Comp. Sci.                75000
58583 Califieri            History                   62000
76543 Singh                Finance                   80000
76766 Crick                Biology                   72000
83821 Brandt               Comp. Sci.                92000

ID    NAME                 DEPT_NAME                SALARY
----- -------------------- -------------------- ----------
98345 Kim                  Elec. Eng.                80000
12    Farhan               Comp. Sci.                90000

13 rows selected.
```

Problems faced :

Syntax Errors: SQL queries must have some strict syntax requirements. Incorrectly worded queries might result in syntax problems and as a beginner I did struggle to get the syntax properly.

Understanding Database Structure: as I was not so familiar with the structure of the database I am working with, it might be difficult to determine which tables exist, what columns they include, and how they relate to one another.

Table Creation: Creating tables with the right structure and data types were a bit tricky. Defining primary keys, **Foreign keys**, and constraints correctly is essential.
Finding out and Declaring Foreign seems hard to me. And command line of declaring foreign key is also.

Connect Table : to connect multiple table command is a bit difficult to me.